

Python Graphs & Charts - Theory and Visual Guide

Simple to Moderate Level • Dataset + Reason + Interpretation + Graph

No Python program/code included

This guide explains the most common charts used in Python data visualization. Each section uses a small, appropriate dataset and explains why that chart is suitable, what it helps you understand, and where it is commonly used.

1. Line Graph

Best for: Changes or trends over time

Sample dataset

Month	Sales
Jan	120
Feb	150
Mar	170
Apr	160
May	200
Jun	230

Why this chart is suitable

The x-axis is an ordered time sequence. Connecting values makes the direction of change easy to see.

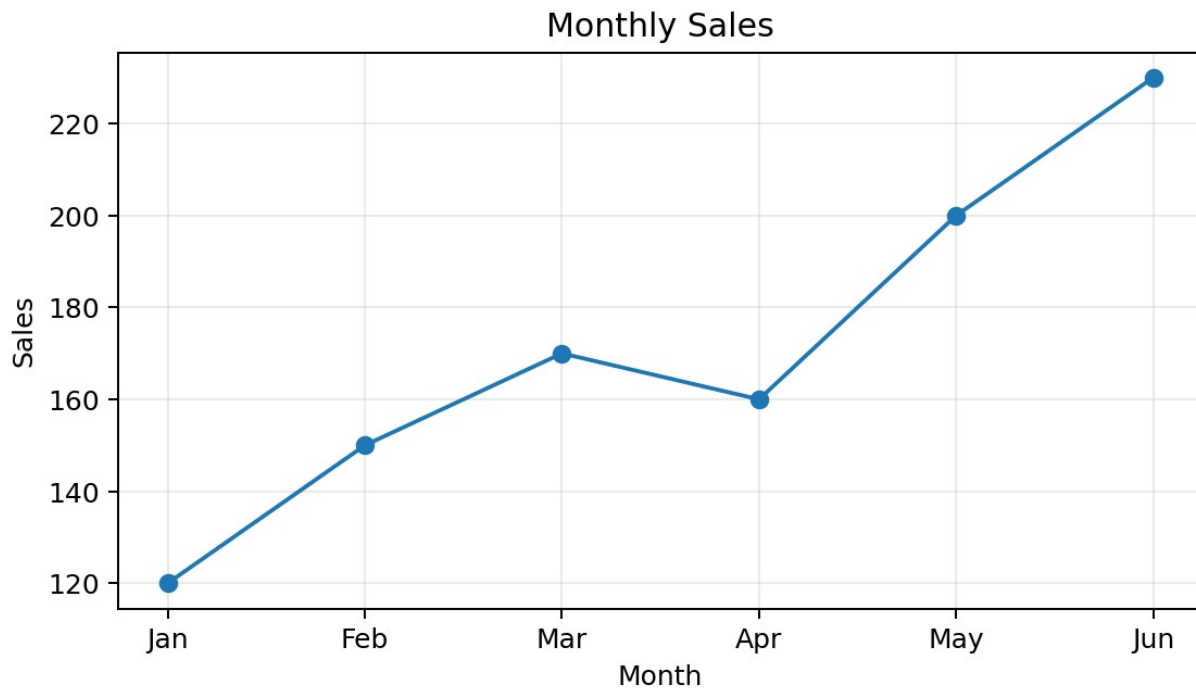


Figure 1: Line Graph

How to interpret it

Sales generally rise from January to June, with a small dip in April.

Most suited for

- Monthly sales
- Daily temperature
- Stock prices
- Yearly population
- Website traffic over time
- Performance across semesters

Key idea: Line chart = trend over time.

2. Bar Chart

Best for: Comparing separate categories

Sample dataset

Student	Marks
Amit	78
Sara	92
John	65
Neha	88
Ali	75

Why this chart is suitable

Each student is a separate category, and the goal is to compare the size of one value across those categories.

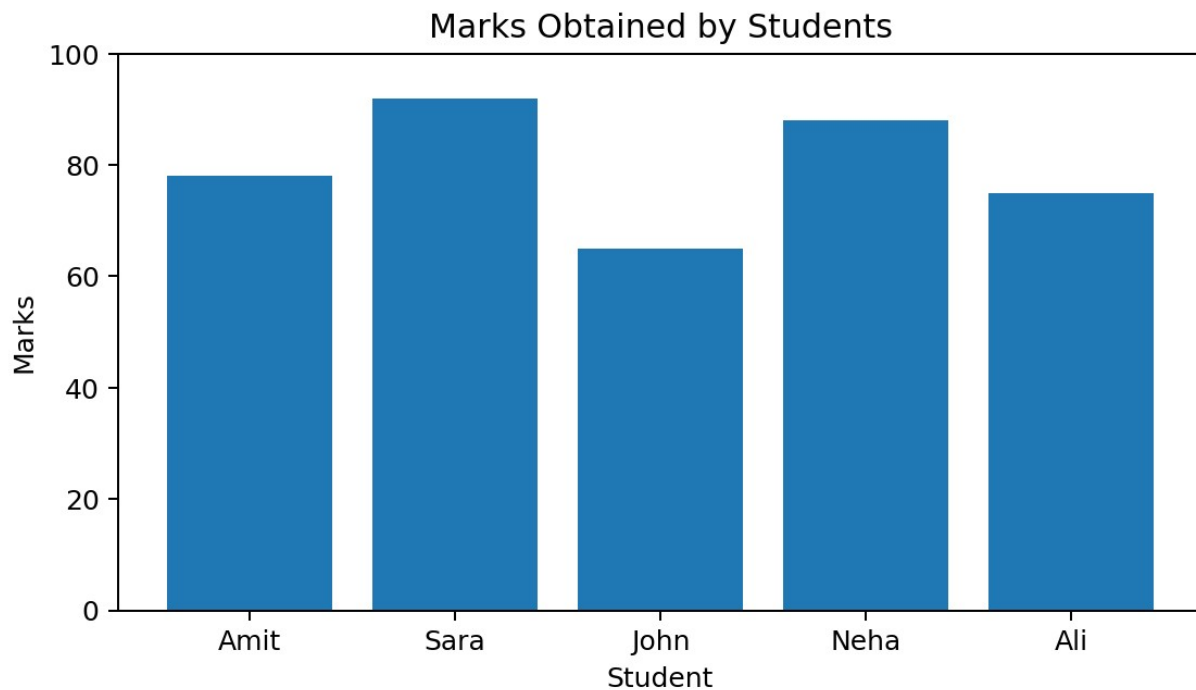


Figure 2: Bar Chart

How to interpret it

Sara has the highest marks and John the lowest. The bar lengths make these differences easy to compare.

Most suited for

- Marks of students
- Sales of different products
- Population of different cities
- Profit by company
- Employees by department

Key idea: Bar chart = compare categories.

3. Horizontal Bar Chart

Best for: Comparing categories when labels are long

Sample dataset

Programming Language	Students
Python	120
Java	90

C++	70
JavaScript	100
R	40

Why this chart is suitable

It provides the same comparison as a normal bar chart but gives more horizontal space for category names.

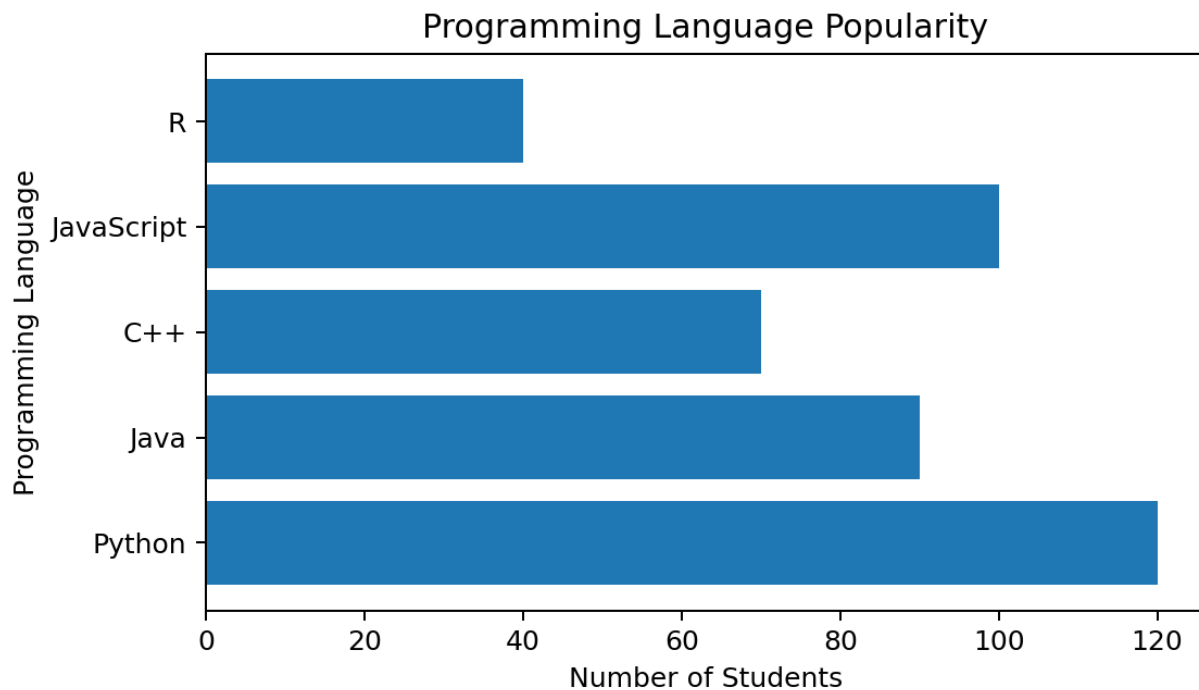


Figure 3: Horizontal Bar Chart

How to interpret it

Python has the largest number of students, while R has the smallest.

Most suited for

- Long category names
- Rankings
- Many categories
- Survey responses

Key idea: Horizontal bar = category comparison with easier-to-read labels.

4. Pie Chart

Best for: Showing parts of one whole

Sample dataset

Department	Budget Share
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Marketing	30
IT	25
HR	15
Operations	20
Research	10

Why this chart is suitable

All values represent shares of one total budget. A pie chart shows how that whole is divided among a small number of categories.

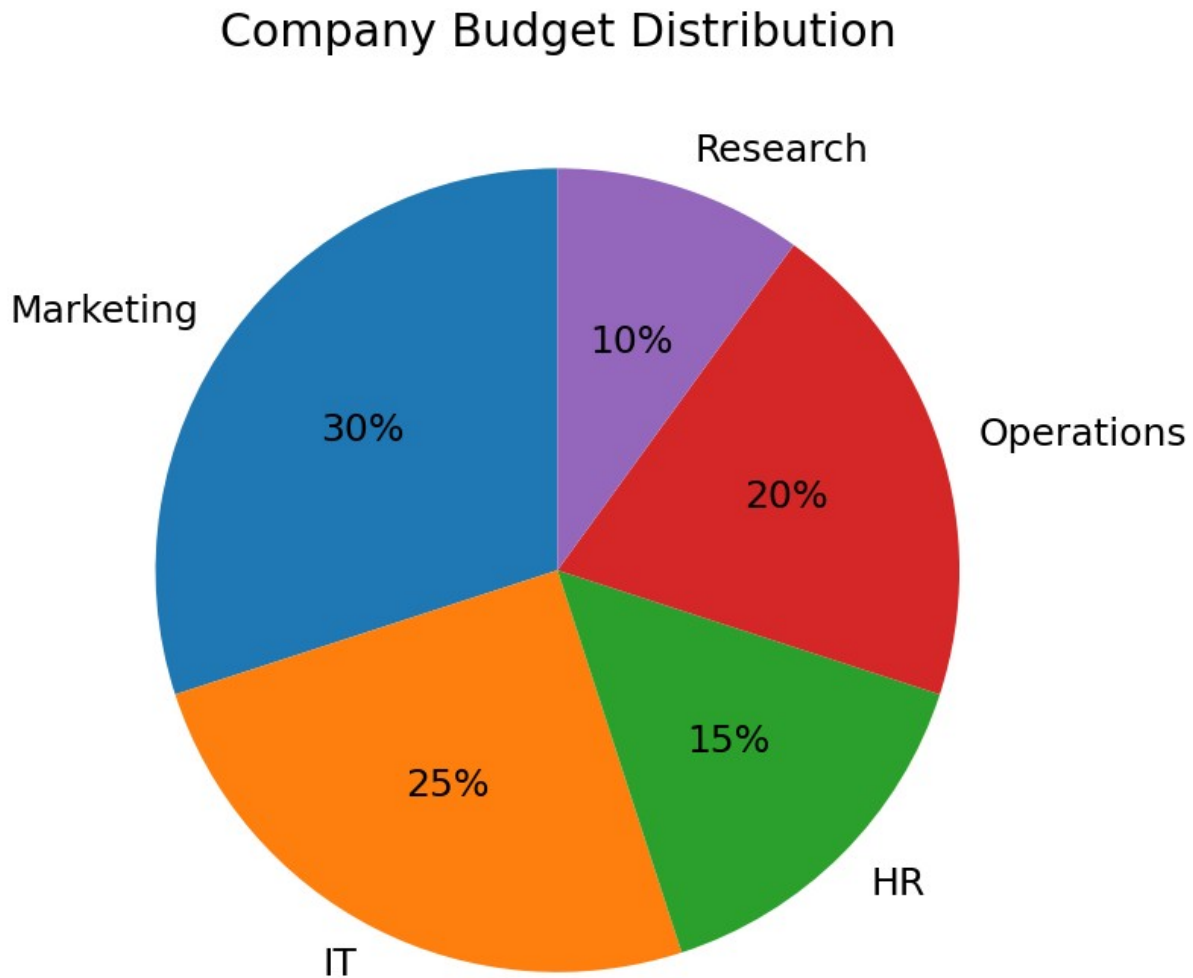


Figure 4: Pie Chart

How to interpret it

Marketing has the largest share at 30%, while Research has the smallest at 10%.

Most suited for

- Budget distribution
- Market share

- Expense distribution
- Small survey category shares

Key idea: Pie chart = parts of a whole. Avoid it when there are too many slices.

5. Histogram

Best for: Understanding the distribution of numerical data

Sample dataset

Marks (30 students)
45, 52, 56, 58, 61, 62, 63, 65, 67, 68, 69, 70, 71, 72, 73
74, 75, 76, 77, 78, 79, 80, 82, 83, 85, 86, 88, 90, 92, 95

Why this chart is suitable

The question is not who scored a particular mark, but how frequently marks fall into ranges. Histograms group continuous numerical data into bins.

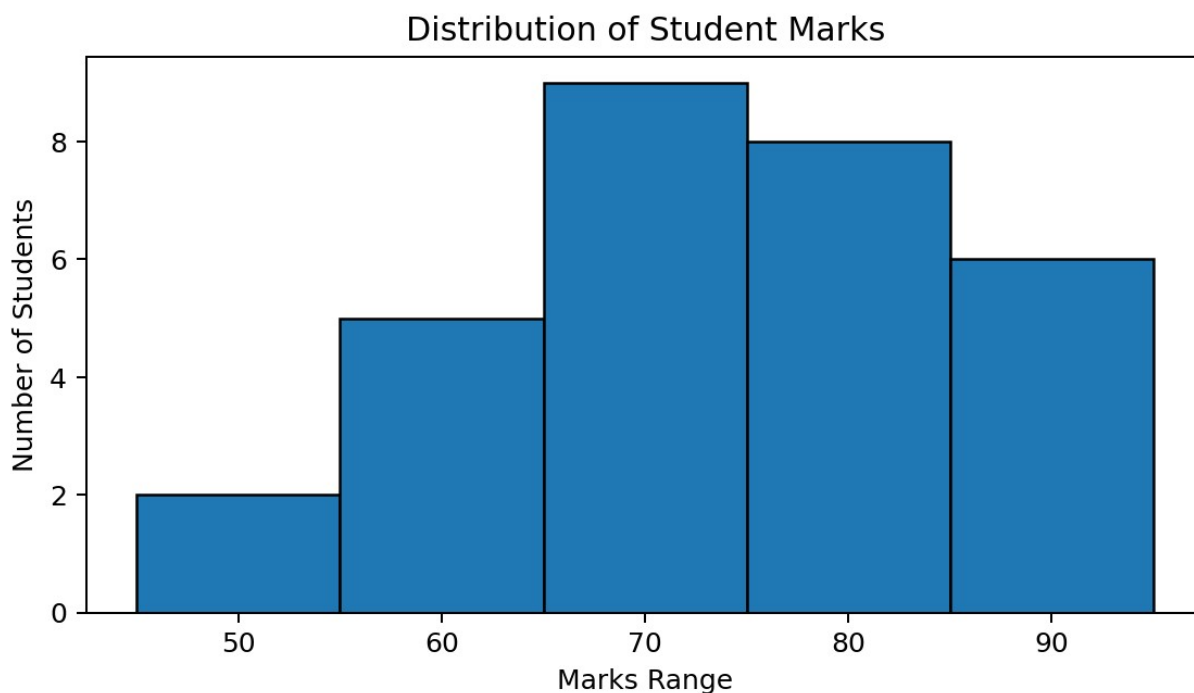


Figure 5: Histogram

How to interpret it

Most scores are concentrated in the middle-to-higher ranges, with fewer scores at the extremes.

Most suited for

- Exam-score distribution
- Age distribution
- Salary distribution

- Purchase amounts
- Height/weight measurements
- Delivery times

Key idea: Histogram = distribution of numerical values.

6. Scatter Plot

Best for: Finding a relationship between two numerical variables

Sample dataset

Study Hours	Marks
1	45
2	50
3	58
4	65
5	70
6	78
7	82
8	88
9	92

Why this chart is suitable

Both variables are numerical, and each student can be represented as one point. The pattern of points helps reveal whether the variables move together.

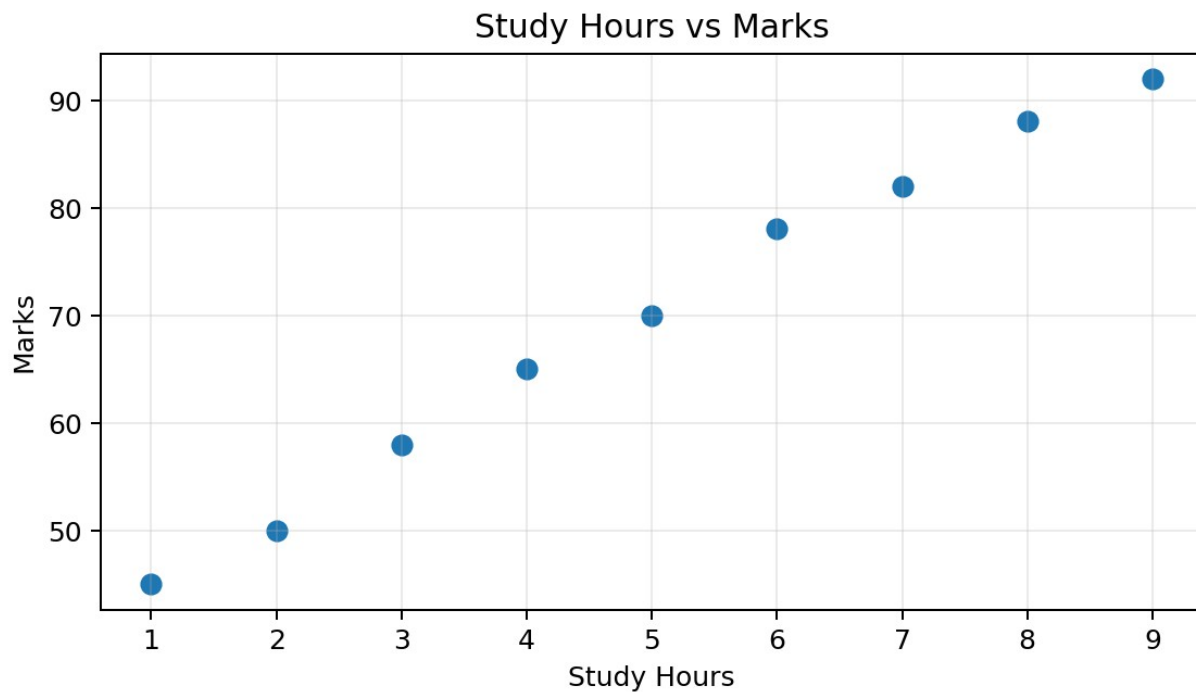


Figure 6: Scatter Plot

How to interpret it

The points rise from left to right, suggesting a positive relationship: students who study more hours tend to have higher marks.

Most suited for

- Study hours vs marks
- Advertising spend vs sales
- Height vs weight
- Experience vs salary
- House size vs price

Key idea: Scatter plot = relationship between two numerical variables.

7. Box Plot

Best for: Viewing spread, median and possible outliers

Sample dataset

Employee Salary
28000
30000
31000
32000
33000
34000
35000
36000
37000
38000
40000
42000
45000
47000
50000
90000

Why this chart is suitable

A box plot summarizes the distribution compactly and is especially good for spotting unusually high or low values.

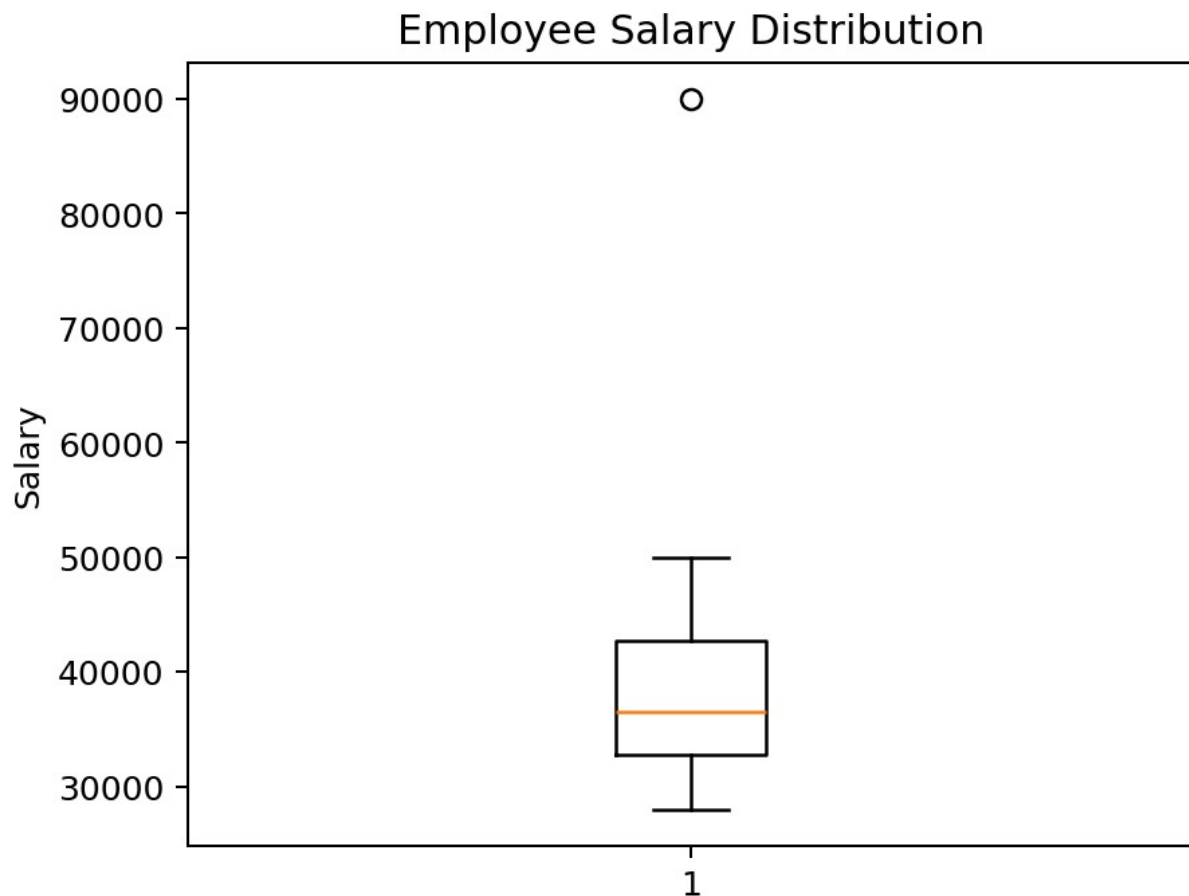


Figure 7: Box Plot

How to interpret it

Most salaries lie between about 28,000 and 50,000, while 90,000 appears as a clear high outlier.

Most suited for

- Outlier detection
- Salary analysis
- Exam-score spread
- Unusual transaction values
- Quality-control measurements

Key idea: Box plot = spread + median + outliers.

8. Multiple Box Plots

Best for: Comparing distributions between groups

Sample dataset

Class A	Class B	Class C
65	50	75
70	55	78

72	60	80
75	65	82
78	70	84
80	75	86
82	80	88
85	85	90

Why this chart is suitable

A bar chart might compare only averages. Multiple box plots compare median, spread and range for several groups at once.

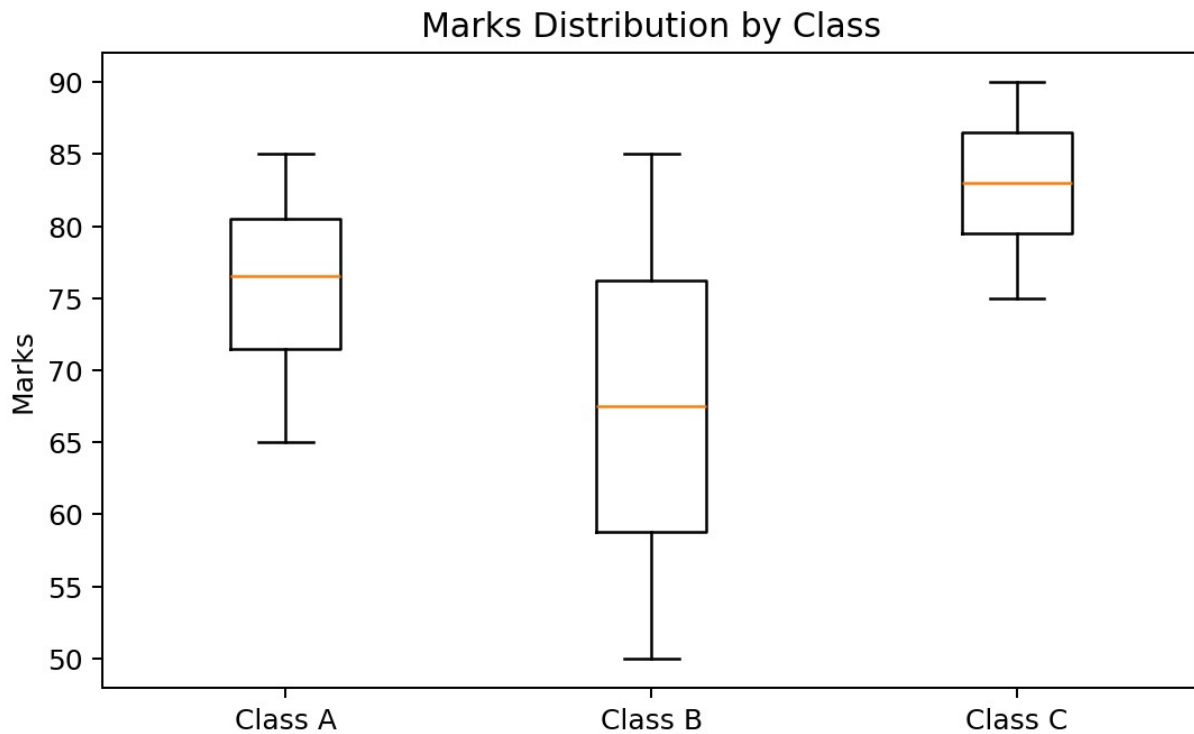


Figure 8: Multiple Box Plots

How to interpret it

Class C is generally higher and more tightly grouped; Class B has a wider spread and lower central values.

Most suited for

- Comparing classes
- Comparing departments
- Comparing treatment groups
- Comparing machine/process performance

Key idea: Multiple box plots = compare entire distributions, not just averages.

9. Area Chart

Best for: Showing trend while emphasizing magnitude

Sample dataset

Month	Visitors
Jan	1000
Feb	1400
Mar	1800
Apr	1600
May	2200
Jun	2700

Why this chart is suitable

Like a line graph it shows change over time, but the filled area makes the overall volume or magnitude more visually prominent.

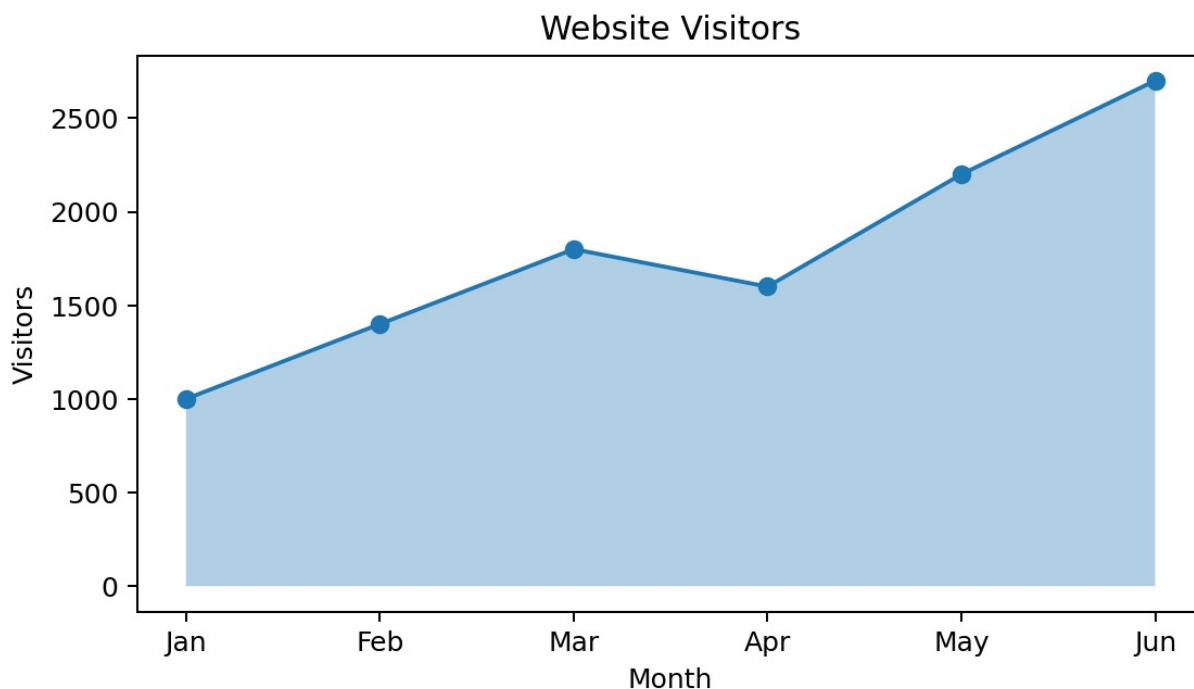


Figure 9: Area Chart

How to interpret it

Website traffic generally grows, with a temporary decline in April followed by stronger growth.

Most suited for

- Website visitors
- Revenue growth
- Electricity consumption
- Population growth
- Production volume over time

Key idea: Area chart = trend + magnitude.

10. Grouped Bar Chart

Best for: Comparing multiple categories inside each group

Sample dataset

Quarter	Product A	Product B
Q1	120	100
Q2	150	130
Q3	170	160
Q4	200	190

Why this chart is suitable

Each quarter contains two comparable values. Side-by-side bars make Product A vs Product B easy to compare within the same quarter.

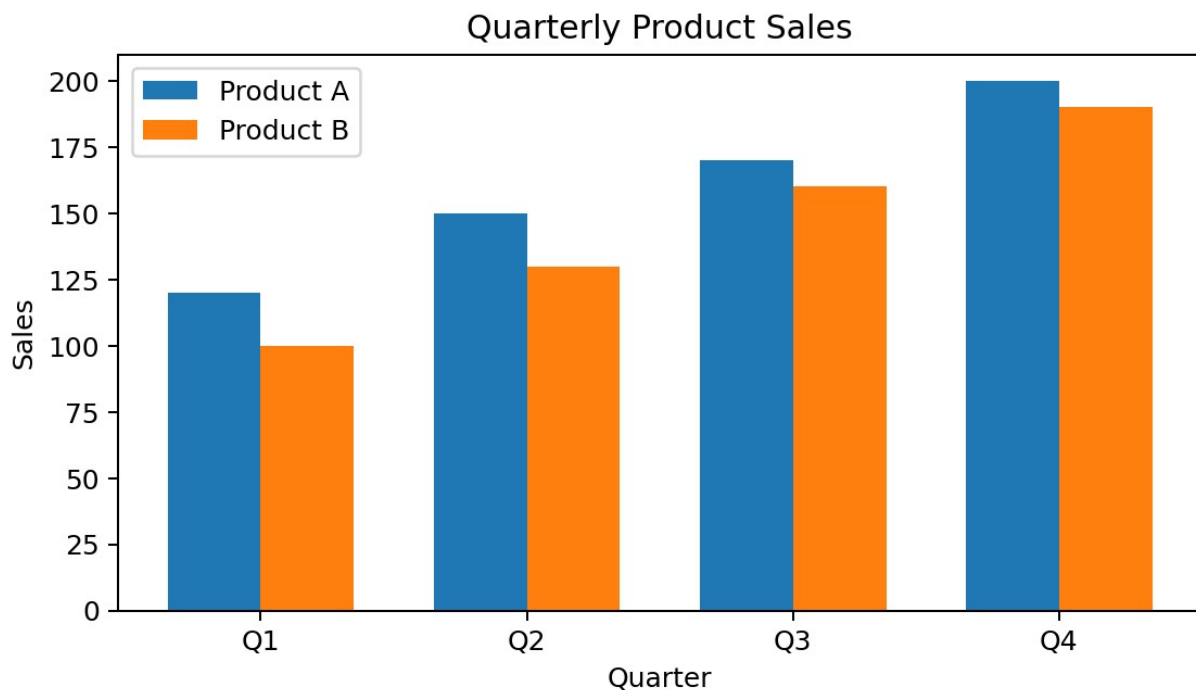


Figure 10: Grouped Bar Chart

How to interpret it

Product A leads Product B in all four quarters, and both products grow over time.

Most suited for

- Product A vs B sales
- Male vs female results by department
- Current year vs previous year
- Brand comparisons across cities

Key idea: Grouped bar = compare multiple series within categories.

11. Stacked Bar Chart

Best for: Showing totals together with their composition

Sample dataset

Department	Male	Female
CS	60	40
IT	50	45
Business	40	55
Arts	30	70

Why this chart is suitable

The height of each full bar shows the total, while stacked segments show how different groups contribute to that total.

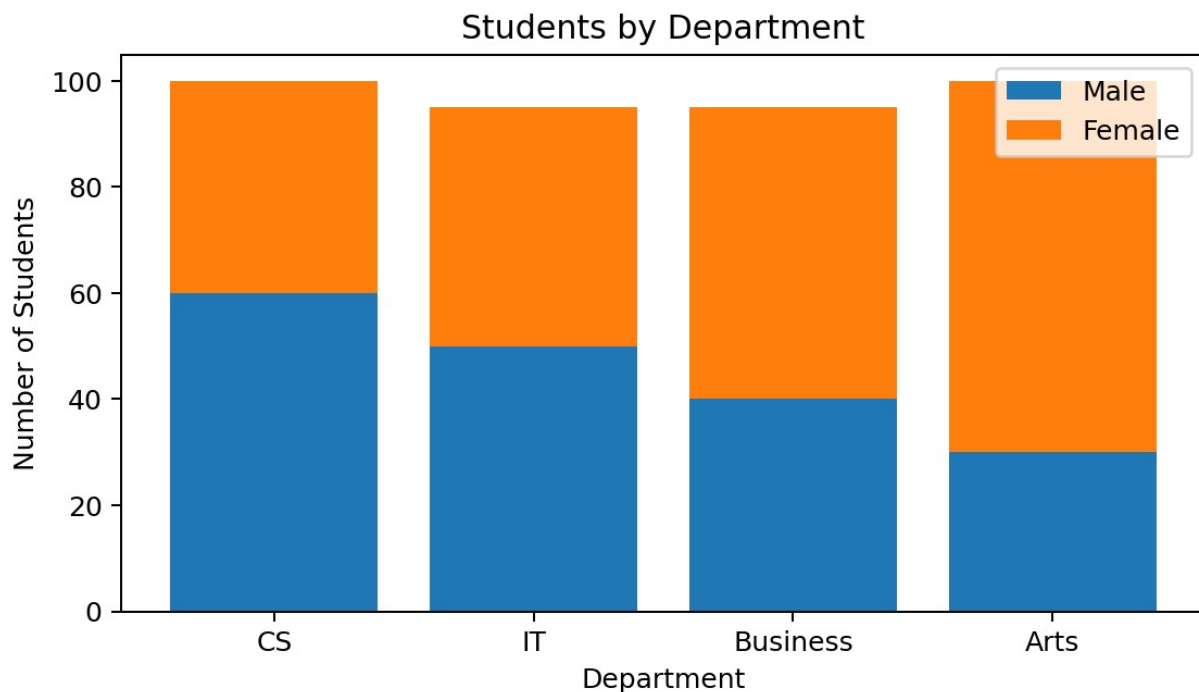


Figure 11: Stacked Bar Chart

How to interpret it

Arts has a larger female share, while CS has a larger male share. Total department sizes can also be compared.

Most suited for

- Gender composition by department
- Expense categories by month
- Product mix by region

- Survey composition across groups

Key idea: Stacked bar = total + parts of total.

12. Multiple Line Graph

Best for: Comparing several trends over time

Sample dataset

Month	Product A	Product B
Jan	100	90
Feb	120	110
Mar	140	150
Apr	160	145
May	180	170
Jun	210	190

Why this chart is suitable

Both products are measured across the same ordered time periods. Separate lines let us compare their trends and crossing points.

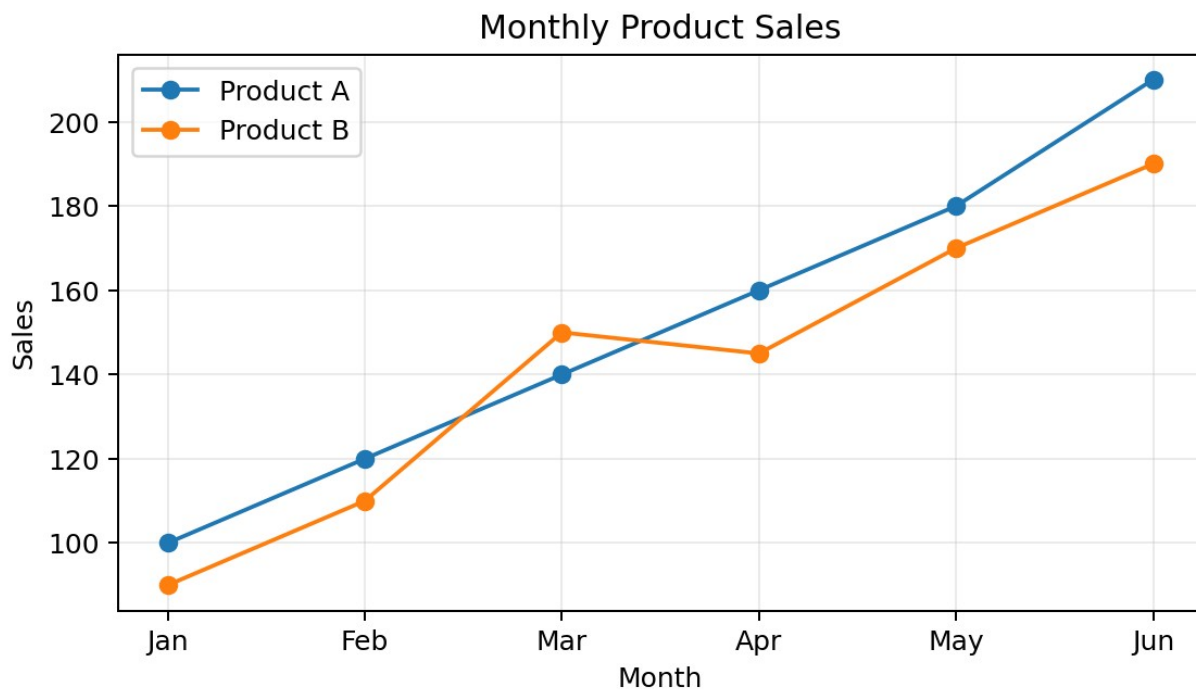


Figure 12: Multiple Line Graph

How to interpret it

Both products trend upward. Product B briefly exceeds Product A in March, while Product A finishes higher.

Most suited for

- Multiple product sales
- Temperatures of different cities
- Different stock series
- Student performance across semesters

Key idea: Multiple line graph = compare trends over the same sequence.

13. Heatmap

Best for: Finding patterns in a matrix of many numerical values

Sample dataset

Day	Morning	Afternoon	Evening
Mon	20	40	60
Tue	25	45	65
Wed	30	50	70
Thu	35	55	75
Fri	40	70	100

Why this chart is suitable

A heatmap turns many numbers into visual intensity, making high and low regions easier to notice than in a plain table.

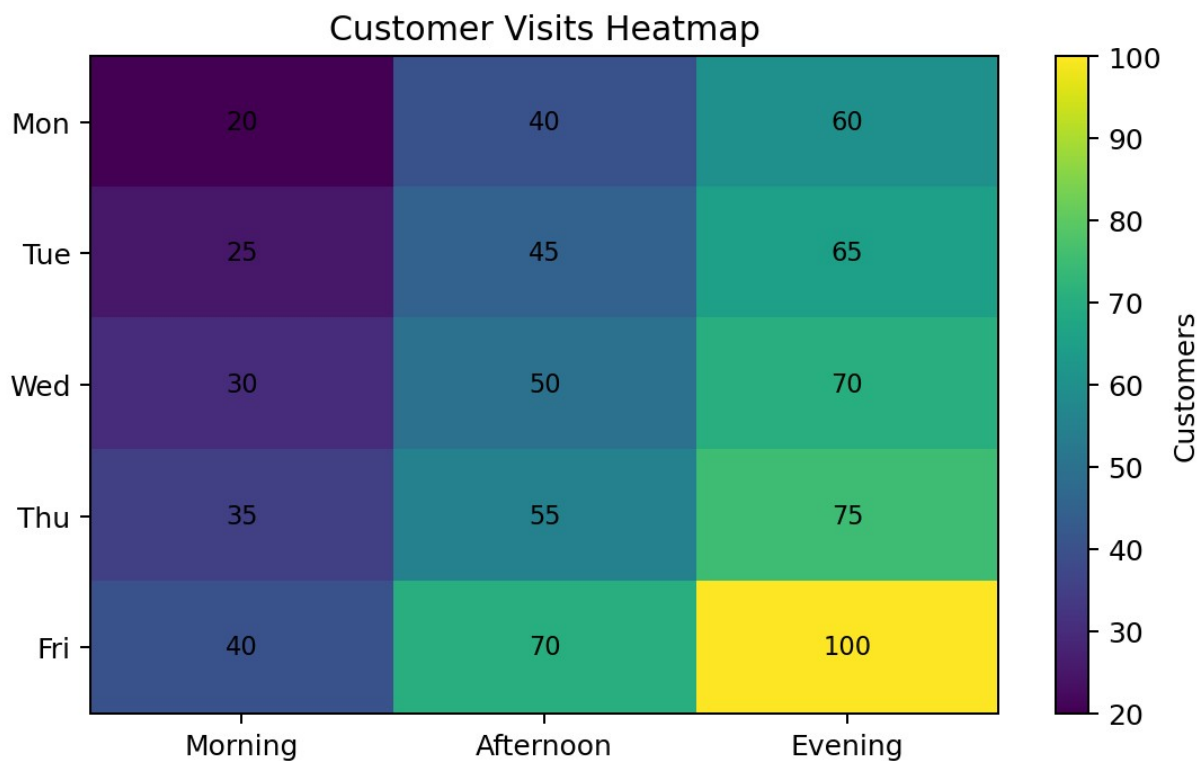


Figure 13: Heatmap

How to interpret it

Customer traffic is strongest in the evenings, especially on Friday, and weakest on Monday morning.

Most suited for

- Correlation matrices
- Sales by day/hour
- Attendance patterns
- Website activity
- Weather grids

Key idea: Heatmap = patterns across a numerical matrix.

14. Stem Plot

Best for: Displaying discrete numerical observations

Sample dataset

Hour	Calls
1	5
2	8
3	4
4	10
5	7
6	12
7	9

Why this chart is suitable

The values occur at specific discrete positions. A stem clearly shows each separate observation without implying a continuous filled region.

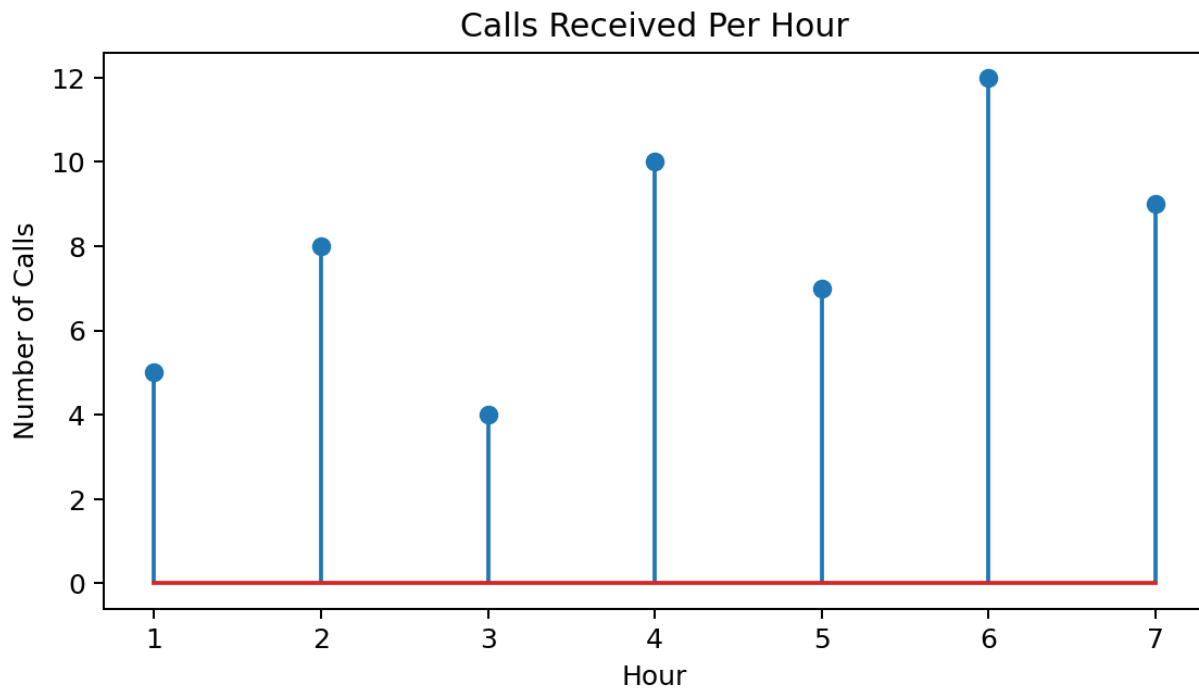


Figure 14: Stem Plot

How to interpret it

Call volume varies hour by hour and reaches its highest point at hour 6.

Most suited for

- Signal processing
- Discrete sequences
- Hourly counts
- Small integer-indexed observations

Key idea: Stem plot = discrete values at specific positions.

15. Step Chart

Best for: Showing values that remain constant until a specific change point

Sample dataset

Units Consumed	Price per Unit
0	2
100	2
200	3
300	3
400	5
500	5

Why this chart is suitable

The price changes only at certain thresholds rather than continuously, so a step pattern matches the way the data behaves.

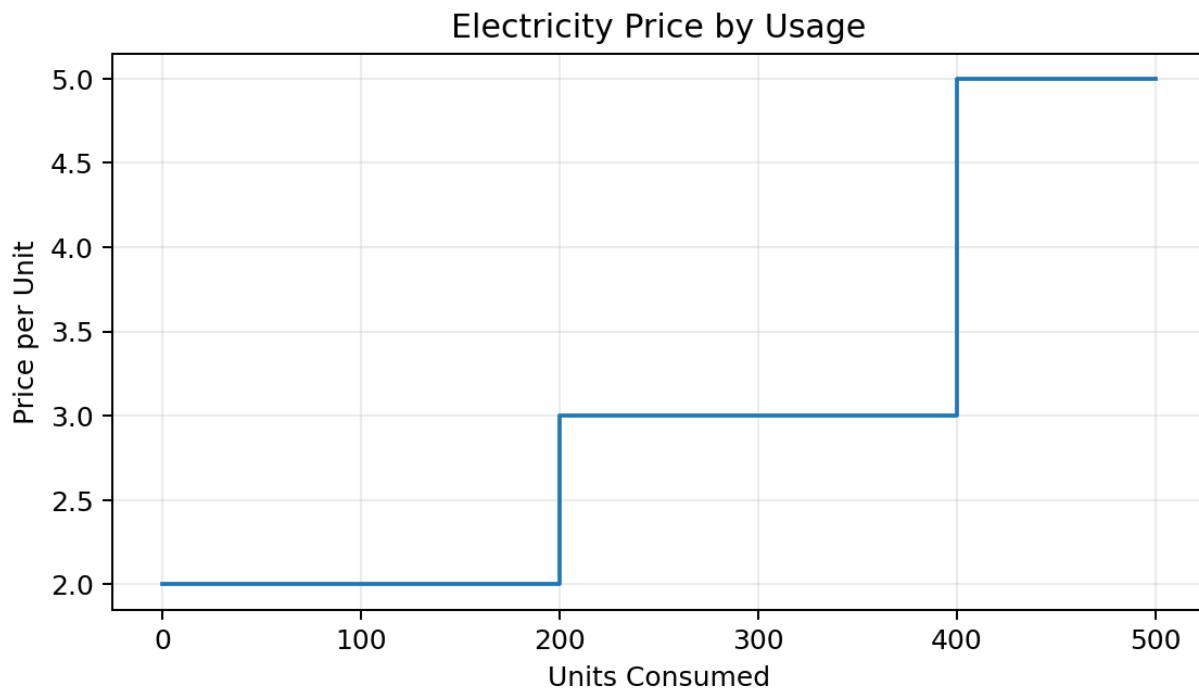


Figure 15: Step Chart

How to interpret it

The tariff stays flat over a range of units and then jumps at defined usage thresholds.

Most suited for

- Electricity tariffs
- Tax brackets
- Inventory levels
- Machine states
- Tiered pricing

Key idea: Step chart = value changes at specific breakpoints.

16. Bubble Chart

Best for: Representing three numerical variables at once

Sample dataset

Company	Revenue	Profit	Employees
A	100	20	100
B	150	35	200
C	80	15	70

D	200	50	300
E	170	40	220

Why this chart is suitable

A bubble chart extends a scatter plot: x and y show two variables while bubble size represents a third.

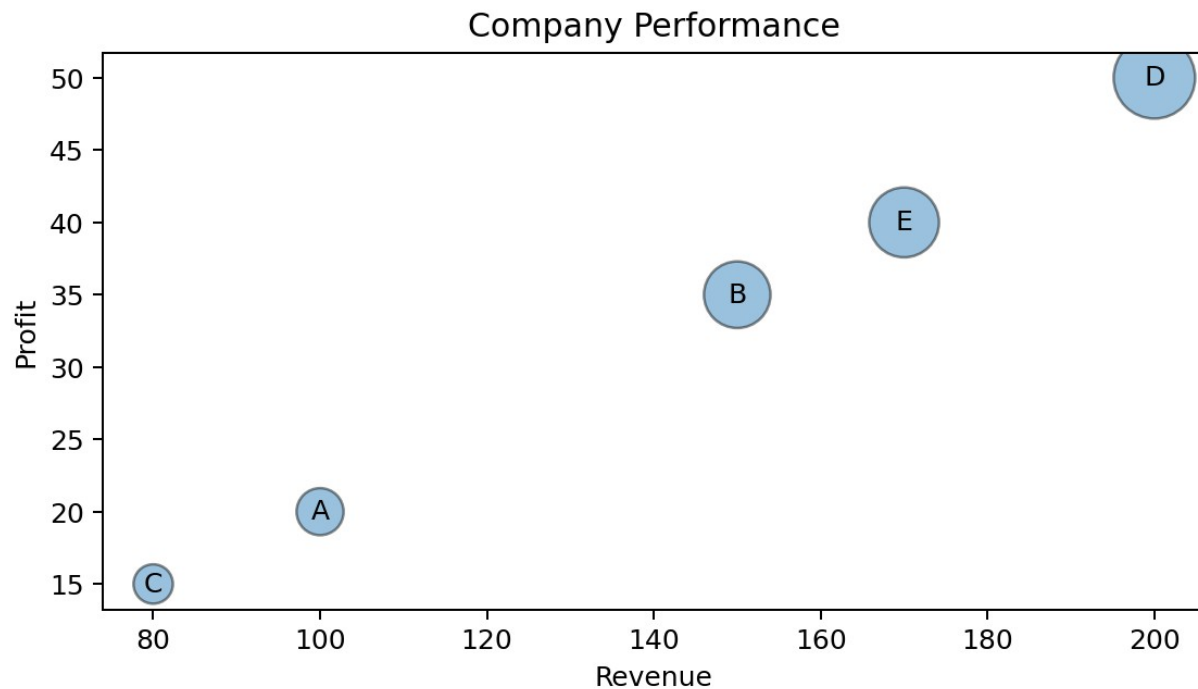


Figure 16: Bubble Chart

How to interpret it

Company D has the highest revenue and profit and also the largest workforce, represented by the largest bubble.

Most suited for

- Company comparisons
- Population vs income with GDP size
- Sales vs profit with market size
- Projects: cost vs benefit with team size

Key idea: Bubble chart = three numerical variables: x, y and bubble size.

17. Error Bar Chart

Best for: Showing averages together with uncertainty or variation

Sample dataset

Experiment	Average	Error
A	50	2

B	55	3
C	53	1.5
D	60	2.5

Why this chart is suitable

A single average can hide variability. Error bars communicate an estimated range around each measurement.

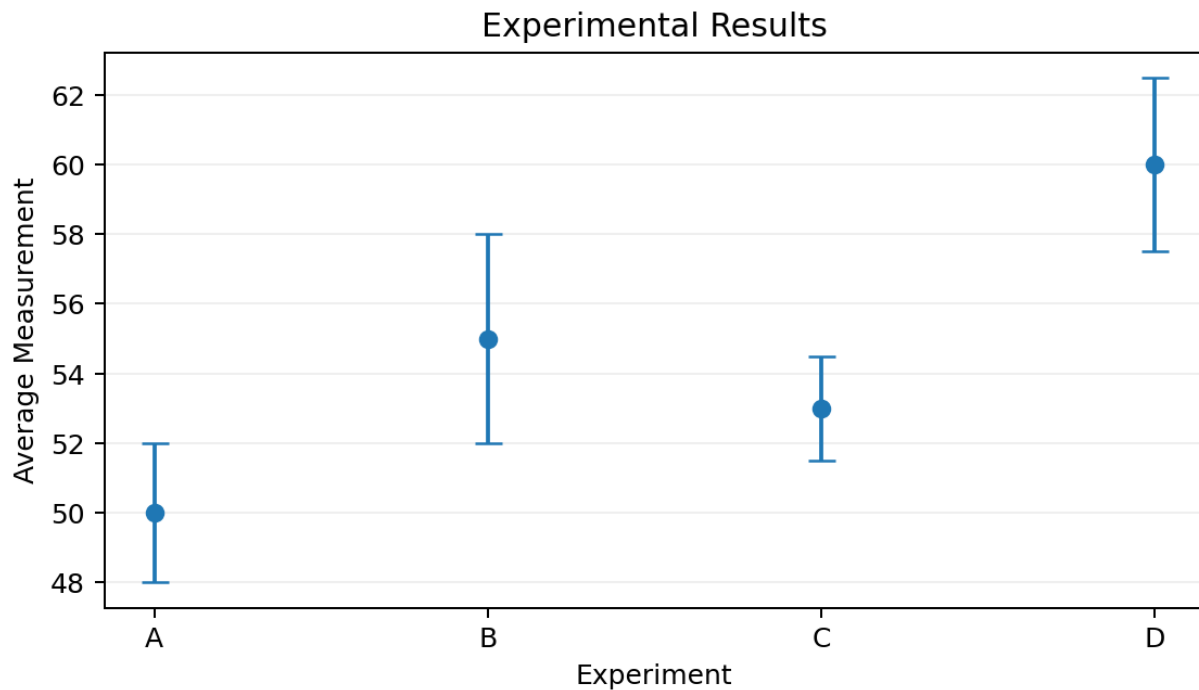


Figure 17: Error Bar Chart

How to interpret it

Experiment D has the highest average, while Experiment B has the widest displayed uncertainty among these examples.

Most suited for

- Scientific experiments
- Confidence intervals
- Repeated measurements
- Laboratory reports
- Statistical summaries

Key idea: Error bar chart = central value + uncertainty/variation.

Quick Chart-Selection Guide

Question you are asking	Best chart
How is something changing over time?	Line chart
Which category is bigger or smaller?	Bar chart
How is one total divided?	Pie chart
How are numerical values distributed?	Histogram
Are two numerical variables related?	Scatter plot
Are there unusual/outlier values?	Box plot
Compare distributions between groups?	Multiple box plots
Compare multiple series inside categories?	Grouped bar
Show total and its composition?	Stacked bar
Compare several trends?	Multiple line
See patterns in a matrix/table?	Heatmap
Show trend plus magnitude?	Area chart
Show three numerical variables?	Bubble chart
Show uncertainty or variation?	Error bar
Show discrete observations?	Stem plot
Values change only at thresholds?	Step chart

The Four Main Purposes of Charts

- 1. Comparison** - Bar chart, grouped bar chart, horizontal bar chart. Example question: Which product has the highest sales?
- 2. Trend** - Line chart, area chart, multiple line chart. Example question: Are sales increasing over time?
- 3. Distribution** - Histogram and box plot. Example question: How are student marks spread out?
- 4. Relationship** - Scatter plot, bubble chart and heatmap/correlation heatmap. Example question: Does advertising expenditure relate to sales?

Easy Memory Trick

TIME → LINE

CATEGORY → BAR

PERCENTAGE / PART OF WHOLE → PIE

DISTRIBUTION → HISTOGRAM

RELATIONSHIP → SCATTER

OUTLIERS → BOX PLOT

MATRIX → HEATMAP

One Dataset Can Require Different Charts

Student	Age	Study Hours	Marks	Attendance
Amit	20	4	65	70
Sara	21	7	88	90
John	19	3	55	60
Neha	22	8	92	95
Ali	20	5	72	80

The same dataset can produce different charts because the chart is chosen according to the question, not merely according to the data file.

Question	Suitable chart
Who scored the highest?	Bar chart
Do longer study hours appear related to higher marks?	Scatter plot
What marks do most students obtain?	Histogram
Are there abnormally high or low marks?	Box plot
Does attendance appear related to marks?	Scatter plot

Core rule: The dataset alone does not determine the visualization. The question you want to answer determines the chart.

Bar Chart vs Histogram - Important Difference

Bar Chart	Histogram
Used for categories	Used for numerical ranges
Compares values across categories	Shows distribution/frequency
Bars are usually separated	Bars normally touch
Example: products and sales	Example: student marks grouped into ranges

End of Study Guide